Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Structure & Interpretation of Computer Programs, Data Structures; *In progress:* Efficient Algorithms & Intractable Problems, Functional Programming & AI-Assisted App Development, Data Engineering

Technical Skills

Languages: Python, Java, SQL, TypeScript

Engineering: FastAPI, React, REST APIs, PostgreSQL, Auth0, AWS S3, Claude API, Apify, pytest, Git, GitHub

Fundamentals: Data Structures, Graph Algorithms, Complexity Analysis, OOP, Functional Programming

ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy

Experience

Guang Labs | Engineering Intern | Remote **May 2026 - Aug 2026**

- Engineered a backend REST API service for an AI video startup that classifies a user's image or video by content type (sports, gaming, talking) using the Gemini API.
- Architected an asynchronous submit-and-poll workflow that processes long videos in the background rather than blocking users (FastAPI, PostgreSQL, Auth0).
- Delivered the service end to end after code review, supporting uploads from **4** sources (local, S3, web links) and validating it with **86** automated tests.
- Developed an in-app notification bell, from the database to the React interface, that alerts users when their video finishes processing or fails, and ensured failures never disrupt processing or send duplicate alerts on retries.

IDX Exchange | Data Science Intern | Remote **Jun 2026 - Aug 2026**

- Engineered a pipeline that predicts California home sale prices from **143K** real-estate records, typically within **8.6%** of the actual price.
- Benchmarked four model types and selected a tuned XGBoost (R^2 of **0.88**), resolving a data leakage bug and cutting a simpler model's error from **29%** to **15%** with a school district feature.
- Analyzed errors by price band to find weak spots: luxury homes (**13%** error vs. **6.5%** mid-priced) and low-sales towns.
- Retrained the final model from scratch and matched results exactly to confirm reproducibility.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Supabase, Apify, Factory Droid* GitHub

- Shaped the idea, features, and vision of a transfer planner for California community college students: code matches courses to official ASSIST agreements (alternatives, grades, agreement years) and plans terms around prerequisites.
- Claude reads transcripts and class schedules, which the student confirms, and answers questions with numbered sources that code checks; general education progress follows each college's Cal-GETC or IGETC list.
- Kept data grounded: Apify scrapes campus pages, a Factory Droid agent repairs broken scrapers under deny-by-default permissions with every patch logged, and guide items must quote their source page.

Resume Tailor | *Python, FastAPI, LaTeX, Claude* GitHub

- Built a tool that tailors a one-page LaTeX resume to a job posting, changing only the bullet text.
- Rejects rewrites that lack a quote from my own records, and checks one-page fit and ATS parsing.

Build Your Own World (Dungeon Game) | *Java* GitHub

- Built a seeded, procedurally generated dungeon game in Java where enemies chase the player along shortest paths (BFS), moving two steps for every one the player takes once in range.
- Designed a quiz-battle system with **100+** CS 61B trivia questions: enemy contact drains health until the player answers correctly, and health packs restore it.

Leadership & Involvement

UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA **Aug 2026 - Present**

- Run exam reviews and write practice worksheets for CS and data science classes for transfer students, and give resume feedback.

UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.